CoopGCN: Cooperative-game-inspired credit signals for long-tail exposure in graph collaborative filtering—a preliminary methods study

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ABSTRACT

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Evidence status: This is a preliminary methods and audit study. Every benchmark cell is one retained run, and the sparse baseline record fails an external calibration check; it is therefore shown only as diagnostic provenance, not as comparative evidence. We formulate ideal cooperative games over graph edges, hyperedges and training interactions, then audit the executable **CoopGCN** checkpoints. The retained code uses deterministic proxies rather than permutation Shapley: centered edge alignment with an explicit tail bonus, affinity-and-tail hyperedge weights, and score-and-tail sample reweighting. Exact Shapley axioms and the local propositions do not apply to those measured proxies. The audit identifies three load-bearing confounds. First, retained LightGCN NDCG@20 is 4–150 times below commonly reported values on Gowalla, Yelp2018 and Amazon-Book. Audit found that a symmetric edge list was reverse-aggregated a second time, double-counting graph messages; corrected reruns are required. Second, cooperative proxy magnitudes are bounded at ±3% (G1), 1% channel mixture (G2), and 1% sample-weight range (G3), while a shared learnable norm scale is unbounded; CoopGCN and LightGCN++ have nearly identical exposure profiles. A frozen-norm ablation is therefore required before attributing exposure to cooperative machinery. Third, only the first 32 neighbours receive non-zero edge-proxy targets, covering at most 27.6% of ML-1M training edges and 56–69% on the three sparse datasets. All three proxies also encode tail preference directly. We provide machine-readable schemas and artifact builders for baseline calibration, multi-seed runs, norm/component ablations, neighbour-cap sweeps, proxy-versus-Shapley validation, corruption, attribution, and runtime. Until those records are complete, this work contributes a scoped architecture, implementation audit and falsifiable experiment plan—not a validated accuracy, exposure, robustness, explanation, or efficiency result.

Highlights

- Sparse baseline calibration fails; retained comparisons are diagnostic only.
- Measured code uses deterministic tail-aware proxies, not MC/TMC Shapley.
- Proxy magnitudes are small; learnable norm scaling is the main identified confound.
- The 32-neighbour cap leaves most ML-1M edges at a neutral proxy target.
- Confirmatory CSV schemas, tables and figure builders are released.

1. Introduction

The retained benchmark initially appeared to show an accuracy–exposure trade-off, but audit changed that interpretation. Sparse LightGCN cells are 4–150 times below published same-dataset references, while CoopGCN and LightGCN++ share nearly identical broad-exposure profiles and a learnable norm scale. The central question is therefore no longer whether CoopGCN wins; it is whether the harness is calibrated and whether cooperative proxies contribute anything beyond norm scaling and explicit tail bonuses.

Graph recommenders' narrow catalogue exposure, sensitivity to noisy data and lack of per-interaction attribution are often treated as separate problems. We investigate whether they share a design limitation inherited from LightGCN: propagation has no explicit representation of how much a particular interaction contributes. Noise resilience and attribution faithfulness are mechanistic hypotheses in this manuscript, not measured findings. Throughout, *credit assignment* means allocating training and propagation influence; it does not mean that the credits are faithful human-facing explanations. We treat explanation faithfulness as a separate, unexecuted evaluation protocol in Section 8.3.

1.1. The LightGCN paradigm and its core trade-off

Graph convolutional networks reshaped collaborative filtering (CF) by modelling user–item interaction histories as bipartite graphs. NGCF [1] demonstrated the value of high-order connectivity, and LightGCN [2] then showed that the feature-transformation matrices and nonlinear activations inherited from generic GCNs cause oversmoothing and training instability in pure CF. Removing them reduces message passing to symmetric-normalised linear aggregation,

$$\mathbf{E}^{(k+1)} = \left(\mathbf{D}^{-\frac{1}{2}}\mathbf{A}\mathbf{D}^{-\frac{1}{2}}\right)\mathbf{E}^{(k)} \tag{1}$$

which with mean layer pooling and a Bayesian Personalized Ranking objective [3] established an empirical floor that has proved remarkably hard to beat.

The three equations defining LightGCN encode three silent assumptions: every neighbour is aggregated with weight $1/\sqrt{(d_u\ d_i)}$, every layer is pooled equally, and every negative is sampled uniformly. The first is our concern here. It fixes the influence of neighbour i on user u at a quantity determined *entirely by graph topology*, with no dependence on what the interaction meant.

Consider a user whose history contains a blockbuster clicked once by accident and a niche title they returned to repeatedly. Equation (1) weights these by degree alone. Worse, because the coefficient is degree-normalised and the blockbuster has the larger degree, the accidental click is down-weighted for a reason that has nothing to do with its being accidental—and the niche title is up-weighted for a reason that has nothing to do with its being meaningful. The ordering happens to be right; the mechanism producing it is blind. Where degree and informativeness diverge, as they do for popular-but-incidental and rare-but-deliberate interactions alike, the model has no parameter with which to tell them apart.

Three consequences follow directly, and they are precisely the three symptoms above. The model does not natively expose a utility-based reliability score for an edge or an intrinsic interaction-level attribution, although post-hoc explainers can be applied. Its fixed weighting also provides no explicit mechanism for discounting a corrupted edge. These are architectural limitations, not evidence that LightGCN cannot be explained or cannot perform well under every corruption process.

1.2. Four structural failure modes of unweighted propagation

A systematic analysis reveals that LightGCN's remaining weaknesses are precisely the capabilities its simplification discarded. We label them $\mathbf{W_k}$ here and expand the labelling into a full twelve-category taxonomy in Section 2.6.

1. **Flat credit and uniform neighbour weighting ($\mathbf{W_1}$, $\mathbf{W_2}$).** Every neighbour contributes with an immutable topological weight. Because that weight is degree-normalised, noisy or irrelevant edges dilute the embedding quality of every node they touch, and the model has no parameter with which to discount them.
2. **Pairwise-only topological bias ($\mathbf{W_5}$).** Bipartite edges capture relations (u,i) but cannot represent higher-order group units such as shopping sessions, item categories or multi-item bundles, which must instead be approximated by cliques.
3. **Popularity-bias amplification and dimensional collapse ($\mathbf{W_3}$, $\mathbf{W_8}$).** Repeated convolution is dominated by the leading eigenvectors of the adjacency matrix. Embeddings collapse toward high-degree items, and tail recall degrades sharply with layer depth.
4. **Vulnerability to noisy and poisoned data ($\mathbf{W_{10}}$).** Because every edge is trusted uniformly, malicious clicks or erroneous interactions ripple unconstrained through k-hop neighbourhoods.

Section 2.6 expands these into a twelve-category taxonomy and argues that the high-severity entries reduce to three root causes.

1.3. The game-theoretic perspective

The natural response is to make the weights learnable, and graph attention does exactly that. Our position is that this is the right instinct applied with the wrong tool, and the reason is worth stating precisely.

Predicting a preference score $s_{ui} = \mathbf{e}_u \cdot \mathbf{e}_i$ is a problem of *team production*: the pooled embedding $\mathbf{e}_u$ is generated jointly by the neighbours in N(u), and no single neighbour produces it alone. Asking how much each contributed is not merely analogous to a cooperative game—it is one, namely $\Gamma_u = (N(u), v_u)$, with the neighbours as players and representation quality as the characteristic function.

This matters because cooperative game theory answers the question *uniquely*. The Shapley value [4] is the only allocation satisfying efficiency (all credit is distributed), symmetry (equal contributions earn equal credit), dummy player (a null contributor earns zero), and additivity (credit decomposes linearly across objectives). Attention weights are not allocations of a characteristic function, so these axioms do not directly apply to them; softmax normalisation should not be confused with Shapley efficiency.

Symmetry assigns equal exact credit only when two players have identical marginal contribution in every coalition. Popular and niche items will rarely meet that condition, and popularity can re-enter through the characteristic function itself. Symmetry therefore does not close the popularity channel. Moreover, deployed propagation retains degree normalisation.

Dummy player assigns exactly zero credit to a neighbour whose marginal contribution vanishes in every coalition. A maliciously injected edge is close to such a player only when its marginal utility is uniformly small; random or targeted injection does not guarantee that condition.

We must immediately qualify what this buys, because the distinction governs every theoretical claim in this paper. The axioms characterise the *allocation* ϕ. They do not automatically characterise the propagation weight we deploy, which is a bounded, strictly positive transform of ϕ (Eq. (3)). A dummy player receives $\phi = 0$ and therefore neutral modulation, not zero weight. Negative credit is required for down-modulation. These statements concern the ideal allocation; the retained implementation uses deterministic proxies with no axiom guarantee (Section 4.2). We therefore call its signals cooperative-game-inspired rather than axiom-preserving or validated explanations.

1.4. Summary of contributions

1. **A scoped taxonomy.** We organise twelve reported graph-CF limitations into three author-assessed themes: flat credit, pairwise-only structure and head-amplifying training choices. Many methods address one or a subset; CoopGCN is designed to combine several responses.
2. **Ideal games with explicit implementation correspondence.** We formulate edge, hyperedge and data-level cooperative games, then state exactly where the retained executable path substitutes deterministic alignment, affinity/tail and score/tail proxies (Section 4.2).
3. **Local theory.** Channel A contains LightGCN propagation at λ=0 and an ideal exact-credit edge has deviation at most $\lambda\varepsilon/(2\tau\sqrt{(d_u d_i)})$ from its zero-credit reference under Proposition 2. Neither proposition is a guarantee for the proxy-based measured model.
4. **Reproducible preliminary protocol.** We pin data sources, specify candidate filtering, hyperedge generation, all retained CoopGCN constants and the single seed, and release a machine-readable 5-dataset/10-configuration record.

5 **Negative evidence audit and confirmatory workflow.** Sparse baseline calibration fails, proxy magnitudes expose a norm-scale confound, and the neighbour cap leaves most ML-1M targets neutral. Schema-controlled CSVs make every required rerun explicit and keep pending rows out of claims.

Objectives. The primary objective is to specify and audit a cooperative-game-inspired graph-CF architecture before treating its checkpoint table as evidence. The audit asks whether baseline calibration and mechanism-identification gates pass, separates ideal Shapley properties from retained proxies, and creates the confirmatory records required for reruns. It does not claim benchmark superiority or component causality.

Roadmap. Section 2 surveys related work and presents the taxonomy; Section 3 develops the axiomatic foundation and the two propositions; Section 4 specifies the architecture; Section 5 analyses cost; Section 6 details the protocol; Section 7 reports the two failed audit gates; and Section 8 concludes with limitations and future work.

2. Related work and systematic taxonomy

2.1. Graph convolutional networks in collaborative filtering

Graph convolution for recommendation descends from spectral GCNs [5], whose symmetric-normalised propagation rule LightGCN ultimately inherits. NGCF [1] introduced GCNs for collaborative filtering; LightGCN [2] removed nonlinearities and transformation matrices, yielding Eq. (1). Subsequent work has largely addressed one weakness at a time. LightGCN++ [6] diagnosed embedding-norm inflexibility and uniform neighbour weighting, proposing learnable scalar norm scaling and node-degree weighting. Its shared norm mechanism is a central confound in the retained audit; Proposition 1 provides only a structural analogy. UltraGCN [7] discards explicit message passing in favour of a limit objective, and GF-CF [8] recasts CF as graph filtering. Contrastive and denoising models form a parallel line: SGL [9] augments the graph by stochastic dropout under an InfoNCE objective; SimGCL [10] shows uniform embedding-space noise outperforms structural augmentation; XSimGCL [11] simplifies this further to a single-pass noise perturbation and attributes its long-tail benefit to more uniformly distributed representations rather than to any notion of interaction value; LightGCL [12] replaces stochastic augmentation with SVD-truncated contrastive augmentation. We adopt a rank-16 LightGCL-style target as a separate regulariser. A recent survey of self-supervised recommendation [13] confirms the pattern across this family. Critically, all of these methods retain topology-determined, value-agnostic edge weights: contrastive learning changes the *distribution* of embeddings but never asks which edge earned its influence.

Two further lines are directly relevant because they attack the same symptoms by other means. *Attention* was the first attempt to make neighbour weights adaptive: GAT [14] assigns learned coefficients via masked self-attention, and this is precisely the heuristic alternative against which our axiomatic construction is positioned (Section 8.3). *Denoising* methods target edge reliability directly—T-CE [15] down-weights high-loss interactions as likely false positives, and RGCF [16] combines hard pruning with soft reliability weights inside GNN message passing. RGCF is the closest prior work in spirit to our $\mathbf{G_1}$: both modulate propagation by an estimated per-edge reliability. The difference is the source of that estimate. RGCF learns reliability from training loss. Ideal G_1 instead defines marginal coalition credit, for which a dummy receives zero credit and neutral modulation under Eq. (3). The retained implementation uses a centered alignment proxy rather than held-out marginal contribution.

Two methodological studies inform our evaluation design. Dacrema et al. [17] showed that many reported neural recommender gains vanish against properly tuned baselines, motivating our explicit warning that tuning traces are missing; and replication analyses of LightGCN [18] document that reported numbers swing materially with normalisation and split choices, motivating the leakage audit of Section 6.1.

2.2. Hypergraph recommender systems

DHCN [19] and HCCF [20] demonstrated that hypergraph convolutional channels capture session-level and category-level group structures, and HPCF [21] added local–global projection. DyHuCoG [22] pioneered preference-aware Monte-Carlo Shapley weighting for hypergraph recommenders, injecting coalition-derived contribution scores as dynamic hyperedge weights.

DyHuCoG is the conceptual predecessor for the ideal hyperedge game. The retained CoopGCN code does not reproduce its member-level Shapley estimator: it uses the deterministic group proxy in Eq. (8), adds an edge proxy and sample reweighting, and includes an SVD contrastive channel. Accordingly, DyHuCoG is a related configuration rather than a controlled G_2-only ablation.

2.3. Explainable AI and Shapley values in machine learning

Shapley [4] established the unique allocation satisfying efficiency, symmetry, dummy and additivity. Lundberg and Lee [23] unified feature attribution via SHAP; Ghorbani and Zou [24] introduced Data Shapley for training-set valuation; and prior work by the present authors applied Shapley values to explaining unsupervised clustering and black-box models [25]. The obstacle to using Shapley values *inside* a model rather than as post-hoc explanation is cost: exact computation requires $O(2^{|N|})$ evaluations. Section 5 states the cost of an ideal permutation estimator and the lower cost of the deterministic proxies actually used. This manuscript does not evaluate explanation faithfulness.

2.4. Explaining graph neural networks

Recommendation-specific explanation also distinguishes internal feature or interaction importance from user-facing reasons and their fidelity [26]. A parallel literature explains *trained* GNNs post hoc. GNNExplainer [27] learns a soft edge/feature mask maximising mutual information with the prediction; PGExplainer [28] amortises that search into a parametric network so explanations generalise inductively; and SubgraphX [29] searches connected subgraphs by Monte-Carlo tree search scored with Shapley values. The taxonomic survey of Yuan et al. [30] organises this space and, more usefully for us, standardises the fidelity/sparsity metrics by which such methods are judged — the metrics our Section 8.3 adopts.

SubgraphX is the closest prior work in machinery, since it too computes Shapley values over graph structure, and the distinction is worth stating precisely because it is easy to miss. SubgraphX is *post hoc and explanatory*: the model is frozen, and Shapley values are computed afterwards to describe a decision already made. CoopGCN is *intrinsic and constitutive*: Shapley values are computed during training and change the propagation weights, so they alter the decision rather than describing it. The two are complementary rather than current experimental comparators. If the internal proxies are later positioned as explanations, GNNExplainer, PGExplainer and SubgraphX are future faithfulness baselines (Section 8.3); no such comparison is reported here.

FastSHAP [31] is mechanistically adjacent because it amortises a credit target into a network. Unlike FastSHAP, our retained target is q_{ui}, not a Shapley value, and we do not evaluate approximation to exact credit. Section 3.4 defines the unmeasured distillation gap. Covert et al. [32] give the unifying account of removal-based explanation that clarifies which axioms survive which approximations, and we follow their framing in separating properties of ϕ from properties of the deployed weight.

2.5. Popularity bias and reliability context

Popularity bias in graph CF is well documented and systematised in the survey of Chen et al. [33]: convolution moves users toward high-degree items, and tail exposure falls as depth grows [20,12]. Mitigations fall into four families, distinguished by *where in the pipeline* they intervene. *Re-ranking* post-processes the finished list, as in the personalised xQuAD variant of Abdollahpouri et al. [34]. *Regularisation* penalises popularity correlation inside the learning-to-rank objective [35]. *Causal* methods model popularity as a confounder and remove its effect by intervention: PDA [36] deconfounds item popularity, while DecRS [37] intervenes on the user representation to curb bias amplification. *Representation-level* methods such as contrastive augmentation act on the embedding distribution [11].

CoopGCN belongs to none of the four, and the distinction is structural rather than terminological. Re-ranking accepts a biased scorer and repairs its output; regularisation and causal deconfounding correct the objective or the inference rule while leaving the aggregation weights topological; contrastive methods reshape the embedding geometry. CoopGCN changes the aggregation coefficient and explicitly inserts tail terms inside its proxies. Whether this improves relevance or merely encodes exposure preference is unresolved; the zero-tail-bonus and frozen-norm ablations are required.

On robustness, standard graph CF trusts all edges equally, an assumption inherited from the confidence-weighted implicit-feedback formulation [38] in which confidence is a function of frequency alone. The threat model matters here: Zugner et al. [39] showed that graph neural networks are vulnerable to *targeted* structural perturbations optimised against the model, which are strictly stronger than the random edge injection protocol pre-specified in Section 8.3. The retained measured record contains no random- or targeted-injection result, so we make no empirical robustness claim. Data Shapley [24] motivates data valuation. The ideal G_3 formulation adapts that idea to interactions, whereas the retained executable path only reweights a deterministic score-and-tail proxy. Whether contrastive augmentation or data valuation provides protection under this protocol remains an open empirical question.

2.6. Systematic taxonomy of limitations

Table 1 catalogues twelve weaknesses of linearised graph CF, separated into representational (W1–W6) and training/data (W7–W12) categories. Reading it vertically rather than row by row, the high-severity entries reduce to three root causes. *Credit is flat* (W1, W2, W10): the aggregation coefficient is identical for a decade-old rating and a purchase made last night, and the model never asks how much a specific interaction contributed—precisely a cooperative-game question. *Structure is pairwise-only* (W5, W9): preference lives in groups, and pairwise edges approximate groups poorly exactly where data is sparse and group evidence is strongest. *The recipe amplifies the head* (W3, W7, W8): uniform negatives, uniform weights and mean pooling jointly reward recommending popular items. Many recent improvements primarily address one or a subset of these factors; CoopGCN is organised to address several simultaneously.

Table 1. Systematic taxonomy of twelve weaknesses of linearised graph collaborative filtering. Severity is our assessment of the magnitude of the reported effect. The final column records how CoopGCN addresses each: $\mathbf{G_1}/\mathbf{G_2}/\mathbf{G_3}$ denote the game level, C the contrastive channel, R the training recipe, and P the evaluation protocol.

#	Weakness	Consequence	Severity	Documented by	Addressed
Representational weaknesses					
W1	Uniform neighbour weighting	A misclick and a passion interaction propagate identically; noisy neighbours dilute every node they touch	High	LightGCN++ [6]	$\mathbf{G_1}$ proxy; effect unmeasured
W2	Embedding-norm inflexibility	Norm carries no information yet cannot be used; layer-0 and layer-k norms are inconsistent, miscalibrating pooling	High	LightGCN++ [6]	$\mathbf{G_1}$, R
W3	Dimensional collapse	Repeated propagation is dominated by leading eigenvectors; the space becomes head-heavy	High	LightGCL [12]	C
W4	Oversmoothing with depth	Performance peaks at 2–3 layers; deep structure is unusable	Medium	LightGCN [2]	C
W5	No beyond-pairwise structure	Sessions, categories and bundles must be approximated by cliques	Medium	HCCF [20]	Partial $\mathbf{G_2}$; <1% sparse footprint
W6	Static, non-temporal embeddings	Recency and preference drift are invisible	Medium	gSASRec [40]	— (future work)
Training and data weaknesses					
W7	BPR with one uniform negative	Under-samples hard negatives; overconfidence inflates head-item scores	High	gSASRec [40]	Not addressed in retained checkpoint
W8	Popularity-bias amplification	Convolution moves users toward popular items; tail recall falls as depth grows	High	HCCF [20], LightGCL [12]	Explicit tail terms; effect unmeasured
W9	Cold start	New nodes have nothing to propagate	Medium	—	— (future work)
W10	No robustness to noisy or poisoned edges	One malicious interaction ripples through k-hop neighbourhoods	Medium	Data Shapley [24]	Not evaluated
W11	Evaluation leakage in the standard protocol	Reported numbers are inflated; popularity baselines close much of the gap	High	[17]	P
W12	Replication fragility	Results swing with normalisation, depth and splits	Medium	[18]	P

3. Theoretical foundations: axiomatic credit assignment

3.1. Cooperative games and the Shapley value

Definition 1 (Cooperative game). *A cooperative game in characteristic function form is a pair Γ = (N, v) where N = {1,,n} is a finite set of players—interaction neighbours, hyperedge members or training samples—and v : 2^N → R is a characteristic value function with v(∅) = 0. The value v(S) measures the utility achieved by coalition S ⊆ N.*

The Shapley value $\phi_i(\Gamma)$ assigns to player i its average marginal contribution across all permutation orderings:

$$\phi_i(\Gamma) = \sum_{S \subseteq \mathcal{N} \setminus \{i\}} \frac{|S|!\,(|\mathcal{N}| - |S| - 1)!}{|\mathcal{N}|!} [v(S \cup \{i\}) - v(S)] \quad (2)$$

Table 2 fixes the notation used throughout.

Table 2. Notation.

Symbol	Definition
G=(U,I,E)	Bipartite graph, built from the training split only
N(u)	Item neighbourhood of user u (players in Γ_u)
d_u, d_i	Node degrees in G
$\mathbf{e}^{(k)}_u$,$\mathbf{e}^{(k)}_i$	Layer-k embeddings, $\in R^d$
$\mathbf{e}_u$,$\mathbf{e}_i$	Pooled embeddings $\Sigma_k \alpha_k \mathbf{e}^{(k)}$
s_{ui}	Preference score $\mathbf{e}_u \cdot \mathbf{e}_i$
H=(V,E_H)	Hypergraph; D_v,D_h degree matrices
Γ=(N,v)	Cooperative game with value function v
ϕ_j	Exact Shapley value in a fixed ideal game
ϕ_j	Conceptual permutation estimate for a fixed restricted game; not used by retained checkpoints
q_{ui}, q_{ui}	Retained deterministic edge proxy and its EMA
r_{ui}, r_{ui}	Retained deterministic data proxy and its EMA
w_{ui}	Deployed Channel-A edge weight
β_h	Retained deterministic hyperedge weight
γ_{ui}	Retained sample weight, ∈[1,1.01]
a_{ui}	Learnable attention logit distilled from q_{ui}
λ,τ	Ideal modulation strength and positive temperature
L,P,M	Neighbour cap and proxy refresh periods

3.2. The four axioms in collaborative filtering

The Shapley allocation is the *only* function satisfying the four axioms of Table 3, each of which maps to an essential requirement in collaborative filtering. This mapping is the conceptual core of the paper: it distinguishes axiomatic *credit allocation* from learned attention. Section 3.4 explains why the deployed weights do not inherit all four properties.

Table 3. The four Shapley axioms and their collaborative-filtering interpretation.

Axiom	Recommender interpretation
Efficiency	Σ_i ∈ N ϕ_i = v(N): all credit for a user's embedding reconstruction is fully accounted for by their historical interactions.
Symmetry	If v(S ∪ {i}) = v(S ∪ {j}) for all S, then $\phi_i = \phi_j$: two interactions of identical utility receive equal Shapley credit. Their credit modulation is equal, although the full propagation weights can still differ through $1/\sqrt{(d_u d_i)}$.
Dummy player	If v(S ∪ {i}) = v(S) for all S, then ϕ_i = 0: a true dummy neighbour receives zero *credit*; a noisy edge need not be a dummy. Note that zero credit maps to neutral modulation 1, not zero *weight*; negative credit is required for down-modulation (Section 3.4).
Additivity	$\phi_i(v_1 + v_2) = \phi_i(v_1) + \phi_i(v_2)$: under a multi-task objective (accuracy + tail diversity) credits decompose linearly across tasks.

3.3. Proposition 1: recovery of LightGCN and LightGCN++

Proposition 1 (Recovery of LightGCN and scalar norm weighting). *Let the edge-level aggregation weight in Channel A be parameterised as*

Then: (i) when λ = 0, Channel A recovers standard LightGCN symmetric propagation exactly; and (ii) when all neighbours in N(u) are interchangeable under symmetric utility, the symmetry axiom forces $\phi_{ui} = \phi_{uj} = c$, reducing the modulation to a uniform scalar factor that generalises LightGCN++ scalar weighting.

Proof. For (i), setting λ = 0 makes the bracketed term equal 1, recovering $w_{ui} = 1/\sqrt{(d_u\, d_i)}$, which is exactly Eq. (1). For (ii), suppose v(S ∪ {i}) = v(S ∪ {j}) for all S ⊆ N(u) {i,j}. The symmetry axiom gives $\phi_i = \phi_j$ for every interchangeable pair, and efficiency then fixes the common value at ϕ_i = v(N)/|N|. Writing c = $\sigma(\phi_i/\tau)$, which is independent of i, Eq. (3) becomes $w_{ui} = \kappa/\sqrt{(d_u\, d_i)}$ with κ = 1+2λ(c-1/2) a uniform modulation, yielding the same scalar-modulated functional family used by LightGCN++. This is a structural analogy, not recovery of every LightGCN++ normalisation and training detail. ■

Channel A therefore contains LightGCN propagation as an exact boundary case and a scalar-modulated family as a symmetric special case. This statement does not claim that the complete tri-channel model recovers either baseline unless the other channels and losses are disabled.

3.4. What the axioms do and do not transfer to the deployed model

Propositions 1 and 2 concern the idealised weight of Eq. (3). The retained model never computes ϕ_{ui}: it distils deterministic q_{ui} through Eq. (14) and deploys Eq. (16). There is therefore no established approximation chain from exact Shapley credit to the measured artifact.

Ideal estimation. Conditional on a fixed capped player set and finite marginal-contribution variance, the standard independent-permutation estimator is unbiased for that *restricted* game with variance O(1/R) [41]; it is not an unbiased estimator of the uncapped neighbourhood game. For complete marginal vectors, efficiency (telescoping) and the dummy property hold for each sampled permutation; additivity holds samplewise if the same permutations are reused across games; symmetry generally holds only in expectation unless sampling is paired. The retained executable configuration does not run this estimator: it uses the proxies stated in Section 4.2, for which none of the four axioms is guaranteed.

Transformation. For $\tau>0$ the map $g(\phi)=1+2\lambda(\sigma(\phi/\tau)-1/2)$ is nondecreasing, and is strictly increasing only when $\lambda>0$. Within a fixed degree context it preserves credit ordering; full propagation weights also depend on $1/\sqrt{(d_u d_i)}$. The transform is neither linear nor credit-preserving: efficiency and additivity do not survive, and zero credit maps to neutral rather than zero or minimum influence. Any axiom claim is therefore about exact ϕ, not $g(\phi)$ or w_{ui}.

Distillation. A future ideal-credit implementation could measure $\delta\_\phi=|\sigma(a/\tau)-\sigma(\phi/\tau)|$. The retained model instead optimises untempered $\sigma(a)$ toward $\sigma($ q) and deploys $\sigma(a/0.5)$. Its relevant proxy gap and temperature distortion are unmeasured, and no Shapley guarantee degrades "gracefully" into this path without first showing that q approximates ϕ.

The honest summary is two-level: the ideal Channel-A construction is Shapley-derived, while the retained measured implementation is only cooperative-game-inspired and uses q_{ui}. No axiom transfers to that proxy without a separate proof.

3.5. Proposition 2: local ideal edge-credit deviation bound

Proposition 2 (Channel-A per-edge deviation under bounded marginal utility). *Let $i^\wedge{*} \in N(u)$ be an uninformative interaction whose marginal embedding-consistency gain is bounded by $|v_u^{cons}(S \cup \{i^\wedge{*}\}) - v_u^{cons}(S)| \le \varepsilon$ for all $S \subseteq N(u)$ $\{i^\wedge{*}\}$, under the consistency utility of Eq. (5). Let $w^{\phi}_{ui^\wedge{*}}$ denote the ideal Channel-A weight of Eq. (3) with exact $\phi_{i^\wedge{*}}$ in place of $\phi_{i^\wedge{*}}$. Then (i) $|\phi_{i^\wedge{*}}| \le \varepsilon$; and (ii) the deviation of $w^{\phi}_{ui^\wedge{*}}$ from its zero-credit reference is at most $\lambda\varepsilon / (2\tau d_u d_{i^\wedge{*}})$. If the Monte-Carlo error is $|\phi_{i^\wedge{*}}-\phi_{i^\wedge{*}}|\le$, the corresponding estimate-driven bound is $\lambda(\varepsilon+)/(2\tau d_u\ d_{i^\wedge{*}})$. Standard LightGCN instead assigns $1/d_u\ d_{i^\wedge{*}}$ independently of credit.*

Proof. Equation (2) expresses $\phi_{i^\wedge{*}}$ as a convex combination of marginal contributions: the permutation coefficients are non-negative and sum to one. Since each marginal contribution is bounded by ε in absolute value, $|\phi_{i^\wedge{*}}| \le \varepsilon$. This establishes part (i). For (ii), write $w^{ref}_{ui^\wedge{*}}$ for the ideal weight at exactly zero credit, namely $1/d_u\ d_{i^\wedge{*}}$ because $\sigma(0)=1/2$. The sigmoid is globally Lipschitz with constant 1/4, so equation $|\ w^{\phi}_{ui^\wedge{*}} - w^{ref}_{ui^\wedge{*}}\ | = 2\lambda\ |\ \sigma(\phi_{i^\wedge{*}}/\tau) - \sigma(0)\ |\ d_u\ d_{i^\wedge{*}} \le \lambda\ \varepsilon 2\tau\ d_u\ d_{i^\wedge{*}}$. equation The estimate-driven bound follows from $|\phi_{i^\wedge{*}}|\le |\phi_{i^\wedge{*}}|+\le\varepsilon+$ and the same Lipschitz argument. For a fixed layer input, the corresponding single-message deviation is this weight difference times $\|\ \mathbf{e}_{i^\wedge{*}}^{(k)}\ \|$. The proposition does not compose this quantity across layers, neighbours, Channels B–C, pruning, or the learned distillation map, and therefore is not an end-to-end embedding or NDCG robustness theorem. ■

Two caveats must be stated explicitly, since the proposition is weaker than a casual reading suggests. First, the bound is on the deviation from the *zero-credit reference weight*, not from zero: zero credit receives exactly the LightGCN topological weight under Eq. (3). Proposition 2 therefore does *not* establish suppression, an end-to-end robustness margin, or a property of the deterministic proxy used in the retained run. Second, the bound is per-layer and per-edge; composing it across K layers and $|N(u)|$ neighbours introduces constants we do not control analytically.

A random injected edge need not have small marginal consistency utility: this requires assumptions about its embedding distribution, covariance and angle to the current coalition mean. We make no corollary from random injection to small ε and do not claim protection against targeted or adaptive attacks.

4. The CoopGCN architecture

4.1. Overview of the tri-channel architecture

CoopGCN processes collaborative filtering interactions through three complementary representations, shown in Fig. 1. **Channel A** (edge-credit GCN) performs local bipartite propagation with the retained G_1 alignment proxy. **Channel B** (group-credit GCN) performs propagation over training-derived hyperedges weighted by the retained G_2 affinity-and-tail proxy. **Channel C** constructs an SVD-truncated low-rank global view of the interaction graph as an InfoNCE target intended to counter dimensional collapse (W3), an effect not isolated here. Outputs are mean-pooled across layers; G_3 reweights training samples; and L_{game} distils only the edge-proxy EMA into attention.

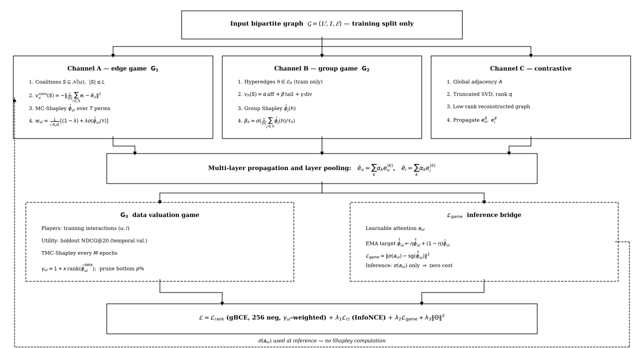

Fig. 1. Exact retained CoopGCN execution path. Channels A, B and G3 use the deterministic proxies of Eqs. (7)–(10); they are motivated by the ideal games but are not permutation Shapley estimators. Channel C supplies a rank-16 SVD contrastive target. The consistency loss distils the edge-proxy EMA into an attention MLP (dashed feedback path).

4.2. Ideal games and retained executable proxies

The cooperative games below define the method's target semantics. The retained checkpoints, however, were produced by deterministic proxies in the released code, not by permutation estimators. Table 4 and the formulas below state that distinction explicitly. Consequently, the measured results evaluate an integrated *game-inspired proxy architecture*; they do not empirically validate exact or Monte-Carlo Shapley allocation.

$\mathbf{G_1}$: edge-level target game and proxy

For each user u, the ideal local game is $\Gamma_u=(N(u),v_u)$. We set $v_u(\varnothing)=0$. Two candidate characteristic functions motivate the design for non-empty S:

Under v^{cons}, exact credit attributes reconstruction of the user embedding, not a particular item's score or rank. Under v^{pref} it attributes the stated coalition utility. We set div(S)=0 for $|S|<2$; otherwise it is the mean pairwise cosine distance, with every norm clamped below by 10^{-8}. The ideal preference utility is not instantiated by the retained run, so α,β,γ have no measured-configuration values.

The retained implementation uses neither expression to enumerate coalitions. For the first L=32 neighbours in training-adjacency order, it computes

$q_{ui}= e_i{}^\wedge{\to}p\ e_u\ -(1)/(|\ N_L(u)|)\Sigma_{j\in\ N_L(u)}\ e_j{}^\wedge{\to}p\ e_u\ +0.05\ 1[i\in\ I_{tail}],$

standardises q_{ui} within the capped neighbourhood when its standard deviation exceeds 10^{-6}, and updates q_{ui}≤ftarrow0.85 q_{ui}+0.15q_{ui}. Negative values represent below-neighbourhood alignment; positive values represent above-neighbourhood alignment. This centered alignment is a deterministic credit proxy, not a Shapley estimate of Eqs.

(5) or (6).

G_2: hyperedge target game and proxy

The ideal group game is Γ_h=(h,v_h), with $v_h(\varnothing)$=0 and a preference-aware affinity/tail/diversity value analogous to Eq. (6). In the retained implementation, no member-level Shapley values are estimated. For a training-derived hyperedge h, embeddings are normalised with an $_2$-norm floor of 10^{-8} and

At each layer the pairwise and hypergraph outputs are mixed as 0.99 e^{pair}+0.01 e^{hyper}. Thus β_h is an affinity-and-tail group proxy, not a group Shapley allocation.

G_3: data-value target game and proxy

The ideal data game takes training interactions as players and validation NDCG@20 after training on coalition S as $v^{ndcg}(S)$. The retained implementation does not retrain coalitions, query validation NDCG, or run TMC. Every M=20 epochs it computes

r_{ui}=σ(s_{ui})× cases1.02,&i;∈ I_{tail},\\0.99,&otherwise;, cases r_{ui}≤ftarrow0.80 r_{ui}+0.20r_{ui},

and sets γ_{ui}=1+0.01qrank(r_{ui}). The released "double argsort" assigns ordinal ranks; equal credits are not average-ranked and can be separated by array order. A bottom-5% mask is calculated, but the retained trainer does not apply it to remove graph edges. The measured configuration therefore uses G3 *reweighting only*; any pruning or denoising effect remains untested.

Table 4. Exact retained CoopGCN configuration represented by the released checkpoints. "Proxy" means that the executable quantity is motivated by a game but is not an exact, MC or TMC Shapley estimator.

Component	Retained setting	Evidence boundary
Initialisation	Seed 42; d=64; 3 layers; embeddings N(0,0.05^2)	One seed only
Optimisation	Adam, lr 10^{-3}, batch 2048, 64 negative IDs generated but weighted BPR uses only the first, up to 1000 epochs, patience 50	No retained search trace
G_1	Eq. (7); L=32; tail bonus 0.05; EMA 0.85; refresh 5; temperature 0.5; λ=0.03; learned norm scale initial 1	Deterministic proxy; legacy permutation parameter 25 is unused
G_2	Eq. (8); affinity coefficient 1; tail coefficient 0.5; temperature 0.5; channel mixture 0.01	Deterministic hyperedge proxy; no member Shapley
G_3	Eq. (10); κ=0.01; EMA 0.80; refresh 20; 5th-percentile mask	Score/tail proxy; mask not applied, no TMC
Contrastive view	SVD rank 16; InfoNCE temperature 0.05; λ_{cl}=0.005	Component effect unmeasured
Attention bridge	MLP 2d→32→1 with ReLU; λ_{game}=0.01; untempered training target, 0.5-temperature propagation	Distillation gap, mismatch and latency unmeasured
Regularisation	λ_{reg}=10^{-4}	Retained configuration archived in code

4.3. Retained objective and proxy consistency regularisation

The retained objective combines ranking, contrastive learning, proxy consistency and L_2 regularisation:

$$\mathcal{L} = \mathcal{L}_{\mathrm{rank}} + \lambda_1 \mathcal{L}_{\mathrm{cl}} + \lambda_2 \mathcal{L}_{\mathrm{game}} + \lambda_3 \|\Theta\|^2 \quad (8)$$

Ranking loss. The retained checkpoints use sample-weighted BPR with one effective negative: although 64 negative IDs are generated per positive, only the first enters

L_{rank}=-| B$|^{-1}$ $\Sigma_{(u,i)\in B}\gamma_{ui}$ σ(s_{ui}-s_{uj_1}).

Thus generalized BCE and multi-negative effects are not part of the measured configuration.

Contrastive loss. Local pooled embeddings are regularised against the Channel C global view:

$$\mathcal{L}_{\mathrm{cl}} = -\sum_{u \in \mathcal{U}} \log \frac{\exp(\mathrm{sim}(\bar{\mathbf{e}}_u, \mathbf{e}_u^{g})/\tau_c)}{\sum_{u'} \exp(\mathrm{sim}(\bar{\mathbf{e}}_u, \bar{\mathbf{e}}_{u'})/\tau_c)} \quad (10)$$

Proxy consistency loss.

$$\mathcal{L}_{\mathrm{game}} = \sum_{(u,i)\in\mathcal{E}} \|\sigma(a_{ui}) - \mathrm{sg}(\sigma(\bar{q}_{ui}))\|^2, \quad \bar{q}_{ui}^{(t)} = 0.85\bar{q}_{ui}^{(t-1)} + 0.15 q_{ui}^{(t)}$$

where a_{ui} is an end-to-end learnable attention logit and sg(·) the stop-gradient operator. The EMA buffer smooths the deterministic proxy across refreshes; it is not an uncertainty estimate. The retained loss is untempered while propagation uses temperature 0.5, an unmeasured train/deploy mismatch.

4.4. Inference with distilled proxy attention

At inference, the retained model evaluates the attention MLP and uses

w_{ui}=()/(√($d_u$$d_i$)) ≤ft[1+2λ(σ($a_{ui}$/0.5)-12)], learnable, initialised at 1.

No credit-proxy refresh is performed on the serving path. The residual attention cost relative to LightGCN is unmeasured, so this is an architectural statement rather than an efficiency result.

5. Computational complexity and retained training algorithm

The ideal Shapley value in Eq. (2) is exponential in the player count; a standard permutation estimator would cost O(TLC_v) per capped local game for T permutations, cap L, and characteristic-function cost C_v. That estimator is not executed by the retained configuration. Its actual periodic costs are:

- edge-proxy refresh: O(| U|Ld) with L=32;
- hyperedge-proxy refresh: O($\Sigma_{h\in E_H}$ $|h|^2$d) for pairwise cosine matrices;
- sample-proxy refresh: one model forward plus O(| E|d) score and rank processing;
- rank-16 sparse SVD: a CPU preprocessing operation whose solver-dependent cost is not reduced to a single closed form here.

Memory includes O(| U|L) for edge EMA/indices, O(| E_H|) for group weights, O(| E|) for sample weights, and model/optimizer state. Wall-clock time and peak memory were not retained; no percentage-overhead claim is made.

Algorithm 1. Retained CoopGCN proxy training path

Input: train graph G, train-derived hyperedges
E_H, periods P=5,M=20
Audit split disjointness; compute train-only degrees and tail mask
Initialise embeddings, attention MLP, q←0, r←1
for epoch t=1,,T
  if t P=1
    Compute edge proxy q_{ui} by Eq. (7); update
    q *// G_1 proxy*
    Compute q_h,β_h by Eq. (8)
     // G_2 proxy
  if t M=1 and t>1
    Compute r_{ui}, r_{ui},γ_{ui} by
    Eq. (10) *// G_3 proxy*
    Compute bottom-5% diagnostic mask; do not alter G
  for minibatch B
    a_{ui}←MLP([e_u| e_i])
    Propagate pairwise and 0.01-weight hypergraph channels;
mean-pool
    layers
    Compute rank-16 SVD contrastive target and
    L by Eq. (11)
    Apply Adam update
  Early-stop on validation NDCG@20 (patience 50)
Inference: use Eq. (16); perform no proxy
refresh

6. Experimental design and methodology

6.1. Split, preprocessing and candidate protocol

The two MovieLens datasets are globally sorted by timestamp (Python's stable sort preserves source-file order for equal timestamps) and split 70/10/20 by interaction count. Gowalla, Yelp2018 and Amazon-Book use the pinned LightGCN snapshot at commit 947ca2b; because those files contain no timestamps, the upstream train split is retained and the upstream test interaction stream is divided 33/67 into validation/test. The latter three splits are therefore *not temporal*. Any validation/test pair already in training is removed; within-split duplicate pairs are otherwise retained.

All graph statistics and tail labels use training edges only. An item is tail when its training degree is at or below the 80th percentile of item degrees. At evaluation, users with at least one ground-truth item are macro-averaged. Each user is ranked against every catalogue item except their training-seen items; validation and test positives are not cross-filtered. PyTorch topk supplies the ranking and no additional tie breaker is imposed. TR@20 averages only over users with at least one tail ground-truth item. Coverage@20 is the number of distinct recommended items divided by the full catalogue size. The released evaluator implements these rules directly.

6.2. Datasets and exact hyperedge construction

Table 5 reports post-loader split counts; the same fields are machine-readable in data/dataset_protocol.csv. Full-source interaction densities are 6.30%, 4.47%, 0.08%, 0.13% and 0.06% in table order; graph construction uses only the listed training count. MovieLens hyperedges group training-present items by genre and retain groups of size at least two. For the three pinned LightGCN datasets, there are no geographic or category metadata in the retained files. The loader instead makes 250 seeded co-occurrence attempts: choose a seed item, sample up to 50 candidate items with replacement, admit candidates sharing at least two training users, stop at size 12, and retain clusters of size at least three. Seed 42 yields the accepted counts and size ranges below; repeated candidate IDs are possible because sampling is with replacement. This proxy construction must not be described as geographic sessions or business/co-purchase categories.

Table 5. Datasets, post-loader splits and retained hyperedges. MovieLens uses global temporal splits; the other datasets use a pinned upstream split without timestamps.

Dataset	Users	Items	Train	Val.	Test	Hyperedges
ML-100K	943	1,682	70,000	10,000	20,000	19 genre groups, size ≥2
ML-1M	6,040	3,706	700,146	100,021	200,042	18 genre groups, size ≥2
Gowalla	29,858	40,981	810,128	71,689	145,553	15 co-occurrence groups, size 3–11
Yelp2018	31,668	38,048	1,237,259	106,968	217,179	41 co-occurrence groups, size 3–6
Amazon-Book	52,643	91,599	2,380,730	199,114	404,264	19 co-occurrence groups, size 3–8

6.3. Retained comparison set

The authoritative measured set is nine baselines plus CoopGCN: **MF** with BPR [3], **NCF** [42], **RecDCL** [43], **LightGCN** [2], **LightGCN++** [6], an in-house **GAT-CF** adaptation, **HCCF** [20], **HPCF** [21], and an in-house implementation of the unpublished **DyHuCoG** preprint [22]. GAT-CF is not a reproduction of a named recommender publication: it uses the same embeddings and topology as LightGCN, with an MLP 2d→32→1 and sigmoid modulation $0.5+0.5\sigma(a_{ui})$. SGL and SimGCL remain related-work methods but are excluded from the retained table because no same-protocol record survives; no claim depends on their omission. Trial-level search traces are unavailable for all models, so every comparison is configuration-specific rather than a fair mechanism-level contest.

6.4. Implementation manifest

The exact retained CoopGCN settings are in Table 4 and paper/retained_run_manifest.yaml; the executable source and checkpoints are archived in the repository. Commit 81da541 is the checkpoint-generation code lineage; later commits revise documentation without claiming new runs. The single global seed is 42. A complete hardware/software manifest and baseline-specific search log were not retained, so Table 6 is not claimed to be independently reproducible to the last decimal from the paper alone.

6.5. Measured metrics

The retained common record contains NDCG@20, TR@20 and Coverage@20 under the candidate protocol above. NDCG uses binary relevance and per-user ideal DCG; TR uses test positives in the training-degree tail stratum; Coverage is a catalogue-level aggregate and does not by itself establish user benefit. Recall and Gini are implemented but omitted because a complete common record was not retained. No corruption metric is part of this study.

6.6. Audit questions and evidence gates

- [**AQ1**] Does the retained benchmark harness reproduce a calibrated LightGCN reference before cross-model comparisons are interpreted?
- [**AQ2**] Are the numerical ranges and coverage of the cooperative proxies large enough, relative to non-cooperative confounds, to motivate causal component claims?

Both gates fail in the retained artifacts (Section 7). Accuracy, exposure, component, corruption, attribution and runtime questions are moved to the confirmatory CSV workflow in experiments/expected/.

6.7. Provenance and statistical scope

The main evidence below is the repository's retained measured record: one executed run per model–dataset cell under the dataset-specific split and unseen-item full-catalogue evaluator described above. Earlier drafts contained a separate table of literature-derived design targets, including projected CoopGCN values of 0.2960, 0.0680 and 0.0480 NDCG@20 on ML-1M, Yelp2018 and Amazon-Book. Executed values were 0.1994, 0.0376 and 0.0235. Because those targets are contradicted by measurement, we remove them, their component table, their random-noise curves and all derived figures from the results section. They remain versioned in the repository as design-history artefacts and must not be cited as findings.

The measured record spans five datasets and ten models, matching Table 6. It contains NDCG@20, TR@20 and Coverage@20 but not a complete common-protocol Recall@20 or Gini record. We do not fill missing cells from publications with different splits. Each value is a single point estimate: no standard deviation, confidence interval or significance test is available, and we make no claim of statistical significance or definitive superiority. Complete tuning traces, including a GAT-CF learning-rate sweep, are also unavailable. The results are consequently preliminary evidence that motivates a multi-seed confirmatory study.

7. Results and discussion

7.1. AQ1: baseline calibration fails

Before cross-model interpretation, we compare the retained LightGCN cells with He et al.'s published same-dataset references [2]. Gowalla is 0.0348 versus 0.1554, Yelp2018 0.0145 versus 0.0530, and Amazon-Book 0.0002 versus 0.0315. Protocol differences preclude expecting exact equality, but ratios of 0.224, 0.274 and 0.006 fail the pre-specified factor-1.5 calibration threshold in experiments/expected/baseline\$_c$alibration.csv. Code audit found a concrete harness fault: edge\$_i$ndex already contains both directions, but every graph model performed a second reverse index\$_a$dd, double-counting every message. The corrected helper now aggregates each directed entry once, but no corrected benchmark has yet been run. AQ1 therefore fails: the retained sparse baseline cells are uncalibrated and cannot support comparative claims.

Table 6 is preserved only as an audit record showing what triggered this conclusion. It is not a leaderboard. No cell is bolded and no relative sparse-data gain is inferred.

Table 6. Uncalibrated single-run audit record. Panel A: NDCG@20; Panel B: TR@20; Panel C: Coverage@20 (%) with item counts reconstructed approximately from rounded percentages. Sparse baseline cells fail calibration and must not be interpreted as comparative evidence.

Model	ML-100K	ML-1M	Gowalla	Yelp2018	Amazon-Book
Panel A: NDCG@20					
CoopGCN (proxy)	0.1826	0.1994	0.1089	0.0376	0.0235
LightGCN++	0.1627	0.2054	0.1092	0.0359	0.0222
GAT-CF	0.1943	0.2096	0.0038	0.0014	0.0002
LightGCN	0.1842	0.2128	0.0348	0.0145	0.0002
HPCF	0.1747	0.2141	0.0278	0.0110	0.0003
HCCF	0.1723	0.2070	0.0291	0.0113	0.0002
DyHuCoG	0.1718	0.2114	0.0338	0.0144	0.0002
RecDCL	0.3103	0.2997	0.0434	0.0109	0.0042
NCF	0.2939	0.2883	0.0444	0.0129	0.0062
MF	0.1467	0.2043	0.0004	0.0006	0.0003
Panel B: Tail Recall TR@20					
CoopGCN (proxy)	0.0106	0.0075	0.0104	0.0006	0.0036
LightGCN++	0.0100	0.0008	0.0116	0.0006	0.0027
GAT-CF	0.0022	0.0001	0.0005	0.0003	0.0000
LightGCN	0.0036	0.0000	0.0000	0.0000	0.0000
HPCF	0.0020	0.0000	0.0000	0.0000	0.0000
HCCF	0.0019	0.0000	0.0000	0.0000	0.0000
DyHuCoG	0.0026	0.0000	0.0000	0.0000	0.0000
RecDCL	0.0000	0.0000	0.0000	0.0000	0.0000
NCF	0.0000	0.0000	0.0000	0.0000	0.0001
MF	0.0052	0.0003	0.0006	0.0006	0.0000
Panel C: Coverage@20 in % (approximate distinct items)					
CoopGCN (proxy)	46.9 (789)	40.7 (1,508)	7.95 (3,258)	5.36 (2,039)	6.47 (5,926)
LightGCN++	46.5 (782)	37.3 (1,382)	8.06 (3,303)	5.01 (1,906)	5.44 (4,983)
GAT-CF	19.4 (326)	10.0 (371)	10.3 (4,221)	0.13 (49)	0.05 (46)
LightGCN	27.8 (468)	9.82 (364)	0.20 (82)	0.35 (133)	0.05 (46)
HPCF	25.7 (432)	9.23 (342)	0.39 (160)	0.48 (183)	0.05 (46)
HCCF	21.2 (357)	8.23 (305)	0.35 (143)	0.25 (95)	0.05 (46)
DyHuCoG	20.5 (345)	9.42 (349)	0.32 (131)	0.35 (133)	0.05 (46)
RecDCL	13.1 (220)	5.42 (201)	0.27 (111)	0.25 (95)	0.05 (46)
NCF	6.54 (110)	4.80 (178)	0.79 (324)	0.84 (320)	0.54 (495)
MF	34.5 (580)	17.1 (634)	6.70 (2,746)	6.82 (2,595)	0.05 (46)

7.2. AQ2: mechanism-magnitude and norm-scale confounding

Table 7 audits the largest numerical influence available to each retained component. The cooperative proxies are deliberately small, whereas the learned norm scale is unbounded. The retained exposure profiles of CoopGCN and LightGCN++ are also close: Coverage@20 is 46.9/40.7/7.95/5.36/6.47 versus 46.5/37.3/8.06/5.01/5.44. LightGCN++ exceeds CoopGCN on Gowalla TR and coverage. The parsimonious unresolved hypothesis is therefore that shared norm scaling, not cooperative credit, drives most of the exposure profile. AQ2 fails until the frozen-=1 row in component\$_a$blation.csv is completed.

Table 7. Retained mechanism magnitude. These are code-level bounds/settings, not measured component effects.

Component	Magnitude	Audit implication
G1 edge proxy	multiplier [0.97,1.03]	±3% maximum
G2 hypergraph	mixture 0.01	1% nominal channel
G3 sample proxy	$\gamma \in [1,1.01]$	≤1% range
Contrastive	λ_{cl}=0.005	effect unmeasured
Proxy loss	λ_{game}=0.01	effect unmeasured
Norm scale	learned, unbounded	primary identified confound

The sparse hypergraph footprint is also tiny. Even before duplicate items are removed, the maximum item slots are 165/40,981 (0.40%) on Gowalla, 246/38,048 (0.65%) on Yelp2018 and 152/91,599 (0.17%) on Amazon-Book, then mixed at 0.01. W5 is therefore only nominally and very partially addressed on these datasets. The values are machine-readable in experiments/expected/hyperedge_footprint.csv.

7.3. Neighbour-cap audit

Only the first L=32 neighbours in training-adjacency order can receive a non-zero q_{ui}. All remaining edges receive a zero target in L_{game}, so they are supervised toward neutral modulation rather than left wholly unsupervised. Table 8 shows that non-zero targets cover at most 27.6% of ML-1M edges and only 56–69% of sparse-data edges. The arbitrary adjacency prefix and the dominant neutral-target class are plausible causes of the ML-1M deficit. Inference additionally sharpens the untempered training logit through $\sigma(a/0.5)$.

Table 8. Training-edge coverage of non-zero G1 proxy targets at L=32. MovieLens values are cardinality upper bounds; sparse values are exact.

Dataset	Mean degree	Targeted fraction	Type
ML-100K	74.2	≤43.1%	upper bound
ML-1M	115.9	≤27.6%	upper bound
Gowalla	27.1	68.8%	exact
Yelp2018	39.1	64.4%	exact
Amazon-Book	45.2	56.4%	exact

7.4. Tail-objective circularity

Every retained proxy encodes tail preference directly: +0.05 in G1, a 0.5 tail-share coefficient in G2, and multipliers 1.02/0.99 in G3. Broader exposure is therefore partly an objective consequence, not evidence that cooperative game structure discovered tail value. G1 then z-standardises within each capped neighbourhood, so the bonus nearly cancels when most capped neighbours are tail and is strongest when tail items are rare. The distribution of capped neighbourhood tail fractions and the zero-tail-bonus ablation are pending in user_degree_diagnostics.csv and component_ablation.csv; no direction or effect size is claimed.

7.5. Protocol audits and status of the retained record

For the three pinned datasets, validation and test are a 33/67 division of the upstream test interaction stream and are not cross-filtered. A validation-only positive can therefore remain in a test candidate set labelled non-relevant, and vice versa, introducing downward false-negative bias when such an item is ranked highly. All configurations call the same released evaluator and training-seen mask, but checkpoint-specific training settings and tuning traces are not sufficiently preserved to establish parity.

Global MovieLens splitting can produce evaluation users with little or no training history. Counts for zero and fewer-than-five training interactions are therefore mandatory calibration outputs; the sparse pinned datasets have zero such users among their retained test users, while MovieLens counts remain pending. Coverage percentages are accompanied by approximate item counts in the audit discussion but are not compared across datasets as method effects.

The confirmatory workflow starts from the schema-controlled CSVs in experiments/expected/. The artifact builder emits LaTeX tables and SVG status figures while rendering incomplete rows as "pending." At this revision, only code-level magnitude, hyperedge-footprint, and partial degree diagnostics are complete. Baseline calibration reruns, norm/component ablations, multi-seed results, L sweeps, proxy-versus-Shapley validation, corruption, attribution and runtime remain pending and are not manuscript results.

7.6. Discussion

The current audit invalidates the earlier empirical interpretation. A working CoopGCN configuration cannot be credited with beating uncalibrated baselines, and a 4–129× exposure difference cannot be attributed to bounded cooperative proxies without freezing the unbounded norm scale and disabling explicit tail terms. The theory remains an ideal design rationale, but no measured quantity currently validates its connection to q_{ui}. The minimum defensible next paper requires calibrated baselines, norm/component ablations and repeated runs; a permutation-Shapley comparison on ML-100K would then determine whether retaining the ideal theory is empirically warranted.

8. Conclusion, limitations and future work

8.1. Conclusion

CoopGCN contributes an ideal cooperative-game formulation and a concrete proxy-based graph-CF architecture. The exact Shapley allocation in the ideal games satisfies the classical axioms; the deterministic edge, hyperedge and sample proxies used by the retained checkpoints do not. Proposition 1 and Proposition 2 concern ideal Channel-A weights only and are not empirical guarantees for the measured implementation.

The retained benchmark record fails sparse-baseline calibration and therefore does not establish a competitive operating point. The code audit further shows that small bounded proxies, explicit tail terms, an unbounded norm scale and a 32-neighbour prefix are confounded. The current contribution is a reproducibly specified method, a negative audit of its retained evidence, and a schema-controlled confirmatory plan—not a validated recommender result.

8.2. Limitations and threats to validity

Calibration and one run. Retained sparse LightGCN values are 0.6–27% of published references under a different protocol. Until the harness is calibrated, these are potentially faulty runs rather than merely uncertain estimates. Every cell also uses seed 42 with no interval or paired test.

Proxy–theory gap. The measured implementation uses Eqs. (7)–(10), not MC/TMC Shapley. Its proxies do not inherit Shapley axioms. Negative edge-proxy values indicate below-neighbourhood alignment and are down-modulated; the G3 quantile mapping largely discards sign and has no dummy-player guarantee.

Component and family confounding. The model simultaneously changes edge credit, hypergraph propagation, sample weights, contrastive view, attention and a learned norm scale. The cooperative ranges are small while the norm scale is unbounded. No measured ablation isolates them.

Tail, cap and hyperedge sensitivity. Tail preference is explicitly encoded in all proxies and is partly cancelled by neighbourhood standardisation in a user-dependent way. At L=32, most ML-1M edges and 31–44% of sparse-data edges receive neutral targets. Sparse hyperedges touch under 1% of catalogue slots at a 0.01 mixture; repeated candidates are possible.

Baseline tuning. Search traces are absent. GAT-CF is an in-house adaptation, and several sparse-data configurations perform near the metric floor. No mechanism-level attention conclusion follows.

Evaluation scope. Candidate ranking excludes training-seen items but uses one cut-off and one split. Score ties have no explicit secondary rule. Coverage is provider/catalogue exposure, not user utility; increasing it may reduce relevance for some users. Only rounded aggregate coverage survives, not per-user exposure distributions.

Validation reuse. The retained G3 proxy does not query validation NDCG, contrary to the ideal data game. Future implementations that do so must use nested valuation/model-selection splits to avoid validation overfitting.

Unvalidated attribution and cost. Internal credits can reveal sensitive historical behaviour and may create unjustified trust if presented as explanations. No deletion, insertion or stability test supports that use. Attention latency, training time, memory and proxy-refresh cost are unmeasured.

8.3. Pre-specified confirmatory work

The execution order is a validity gate. First, rerun LightGCN on the three pinned sparse datasets through the identical evaluator until calibration is within a factor 1.5 of verified references or explain the residual protocol gap. Second, freeze =1 and run the matched component/tail ablations in component_ablation.csv. Third, run at least three seeds on ML-1M and one calibrated sparse dataset. Then sweep $L \in \{16,32,64,128\}$, compare q_{ui} with a 50-permutation ideal estimator on ML-100K, and report training, refresh, memory and serving cost. Corruption and attribution tests follow only if corresponding claims are retained.

All required schemas already exist in experiments/expected/; scripts/build_confirmatory_artifacts.py renders incomplete rows as pending and cannot turn them into claims. None of these are preregistered or completed results.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Ethics and research integrity

This study uses five publicly released recommendation benchmarks and recruits no human participants. The main table contains single-run point estimates with no significance claim. Projection-only leaderboards and synthetic ablation or noise curves are excluded. Internal interaction credits can reveal sensitive historical behaviour; absent faithfulness tests, they are not explanations and must not be shown to users as reasons. Broader catalogue exposure may benefit providers but can reduce relevance or satisfaction for some users, so provider coverage is not treated as an unqualified social benefit.

Declaration of generative AI use

During the preparation of this work the authors used a large language model to assist with language editing and LaTeX formatting. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Data availability

MovieLens-100K and MovieLens-1M are available from GroupLens. Gowalla, Yelp2018 and Amazon-Book are pinned to LightGCN-PyTorch commit 947ca2b3b1d2d3545b114145710cb06c4e57b3d2. Source, checkpoints, data/measured_results.csv, confirmatory schemas, generated pending tables/figures, and artifact builders are available at https://github.com/mouadlouhichi/CoopGCN; checkpoint provenance traces to the 81da541 code lineage.

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